TUTORIAL – 10: Automation & Regression Analysis

Overview & Aim

Aim: Analyze black-box testing, white-box testing, integration testing, and system testing on a sample web application.

Automation Testing is a software testing technique in which test cases are executed automatically using testing tools instead of being executed manually. It is particularly useful for repetitive, stable, and frequently executed test cases.

Regression Testing is performed after fixing bugs or modifying the software to ensure that existing functionalities continue to work correctly and that no new defects have been introduced.

Key Benefits

Benefits of Automation Testing

- Saves testing time and accelerates feedback cycles.
- Reduces human errors in repetitive scenarios.
- Supports repeated, unattended execution.
- Improves overall test coverage.
- Generates faster, consistent reports.

Benefits of Regression Testing

- Verifies that recent code changes do not affect existing features.
- Ensures software stability across consecutive releases and updates.
- Detects unintended side effects of modifications.

Common Automation Testing Tools

- **Selenium** (Web UI Automation)
- **Cypress** (Modern End-to-End Testing)
- **Playwright** (Cross-Browser Web Testing)
- **TestNG** / **JUnit** (Java Testing Frameworks)
- **PyTest** (Python Testing Framework)

Problem Statement

Analyze the modules of the Student Management System and identify which modules are suitable for automation testing. Execute a regression test suite after a software update and prepare an Automation Feasibility Report.

Implementation Details

Parameter	Details
Project Name	Student Management System
Application/Software	Student Management System
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Python
IDE/Tool Used	Visual Studio Code / PyCharm
Input/Test Data	admin/admin123, admin/12345

Source Code

```
# Login Module Verification Script

username = input("Enter Username: ")

password = input("Enter Password: ")

if username == "admin" and password == "admin123":

    print("Login Successful")

    print("Dashboard Opened")

else:
```

```
print("Invalid Username or Password")
```

Module Analysis Table

Module Name	Functionality	Critical (Yes/No)	Frequently Changed (Yes/No)	Suitable for Automation (Yes/No)	Reason
Login	User Authentication	Yes	No	Yes	Frequently executed and stable
Registration	New User Registration	Yes	No	Yes	Validation testing is repetitive
Student Details	Update Student Info	Yes	Yes	No	Frequent business logic changes
Search Student	Search Student Record	Yes	No	Yes	Repeated search operations
Logout	User Logout	No	No	Yes	Simple repetitive functionality

Automation Candidate Identification

Test Case ID	Functionality	Automation Recommended (Yes/No)	Reason	Automation Tool
TC001	Login	Yes	Frequently executed	Selenium
TC002	Registration	Yes	Stable functionality	Selenium
TC003	Search Student	Yes	Repetitive operation	Selenium
TC004	Student Details Update	No	Business logic changes frequently	Manual Testing
TC005	Logout	Yes	Simple repetitive task	Selenium

Detailed Test Case (TC001)

Field	Details
Test Case ID	TC001
Requirement ID	R001

Field	Details
Module Name	Login Module
Feature Under Test	User Authentication
Test Scenario	Login with valid username and password
Test Objective	Verify successful login after software update
Test Type	Regression Testing
Test Technique	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is deployed and Login page is available
Postconditions	User enters Dashboard
Test Environment	Windows 10
Browser/OS	Google Chrome
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful
Status (Pass/Fail)	Pass
Executed By	Student
Execution Date	11/09/2026

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Launch Application	Open URL	Login Page displayed	Login Page displayed	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass
3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login	Click Button	Dashboard Opens	Dashboard Opens	Pass
5	Verify Existing Features	Navigate	Features work correctly	Features working correctly	Pass

Regression Test Suite

Regression Test ID	Module	Scenario	Expected Result	Status
RT001	Login	Valid Login	Login Successful	Pass
RT002	Registration	New User Registration	Registration Successful	Pass
RT003	Search Student	Search Existing Student	Student Record Displayed	Pass
RT004	Logout	Logout	Return to Login Page	Pass

Automation Feasibility Report

Criteria	Observation
Repetitive Test Cases	Yes
Stable Functionality	Yes
High Business Priority	Yes
Frequently Executed	Yes
Suitable Automation Tool	Selenium

Criteria	Observation
Recommendation	Automate Login, Registration, Search, and ...

Regression Testing Summary

Metric	Value
Total Modules Analyzed	5
Automation Candidates	4
Regression Test Cases	4
Modules Excluded	1 (Student Details Update)
Recommendation	Automate stable and repetitive modules; continue manual testing for frequently changing modules

Observations & Remarks

Sr. No.	Observation	Remarks
1	Login module suitable for automation	Yes
2	Registration module suitable for automation	Yes
3	Student Details Update requires manual testing	Business logic changes fre...
4	Regression test cases executed successfully	Pass
5	Existing functionalities unaffected after update	Completed

Output Summary

- **Student Management System Automation Analysis:** Completed
- **Automation Candidates:** Login, Registration, Search Student, Logout
- **Regression Testing:** Completed (Total: 4, Passed: 4, Failed: 0)
- **Overall Result:** Automation Recommended, Regression Testing PASS, Application Stable After Update

Questionnaire Answers

1. **Which testing technique verifies functionality without examining source code?**
    - **Answer:** C) Black-Box Testing
2. **Which testing technique requires knowledge of internal program code?**
    - **Answer:** D) White-Box Testing
3. **Integration Testing is mainly used to verify:**
    - **Answer:** B) Communication between modules
4. **System Testing validates:**
    - **Answer:** C) Complete Application Behavior
5. **Which testing checks execution paths and branch coverage?**
    - **Answer:** B) White-Box Testing

Conclusion / Outcome

- Understood the core concepts of Automation Testing and Regression Testing.
- Successfully identified modules suitable for automation based on stability, execution frequency, and business importance.
- Prepared a detailed Automation Feasibility Report and Regression Test Suite.
- Executed regression test cases to verify that existing application functionalities remained unaffected after updates.
- Demonstrated how automation testing improves efficiency while regression testing guarantees structural stability over time.